

# Triadic Quantum Closure III

## Bridge Closure, Representation Consistency, and a Discrete Gauge Test

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### Abstract

This report evaluates and extends Triadic Quantum Closure III. The note’s central advance is methodological: twelve descriptive roles are separated from physical degrees of freedom; discrete–continuous restriction and reconstruction are separated from history amplitudes and readout; and complex phase is tested against a paired-real control. We reproduce the finite bridge theorem, the phase-interference control, and the discrete gauge calculation. We add a triangulated mesh benchmark with flat Regge geometry and a compact  $U(1)$  connection. The Wilson action converges to its continuum target with error decreasing from  $3.25 \times 10^{-4}$  on a  $2 \times 2$  mesh to below  $5 \times 10^{-9}$  on a  $32 \times 32$  mesh, while gauge-transformed triangle fluxes change by at most  $8.9 \times 10^{-16}$ . These are exact or synthetic verification results, not evidence that the proposed bridge derives quantum mechanics or physical Einstein–Maxwell dynamics. The strongest current conclusion is that TQC III is a precise, falsifiable research framework whose decisive test remains a non-flat, gauge-coupled, refinement-controlled model with held-out observables.

## 1 Question and scope

The project asks whether a triadic relational grammar can provide a useful bridge between discrete geometric/gauge data and a history calculus capable of interference. The present report treats the proposal as mathematics and computational physics. It does not promote a slot count, diagram, or synthetic numerical agreement into a physical law.

## 2 The twelvefold grammar

Let

$$\mathcal{I} = \{1, 2, 3\} \times \{G, E\} \times \{D, C\}, \quad |\mathcal{I}| = 12.$$

The first coordinate marks position inside a triad;  $G/E$  marks gravity/electromagnetism;  $D/C$  marks discrete/continuous representation. The correct state space is typed:

$$\mathcal{X} = \mathcal{X}_{GD} \times \mathcal{X}_{ED} \times \mathcal{X}_{GC} \times \mathcal{X}_{EC},$$

not automatically  $\mathbb{R}^{12}$  or  $\mathbb{C}^{12}$ . A restriction  $R_h : \mathcal{X}_C \rightarrow \mathcal{X}_{D,h}$  and reconstruction  $W_h : \mathcal{X}_{D,h} \rightarrow \mathcal{X}_C$  can satisfy  $R_h W_h = I$  on a discrete range while  $W_h R_h = \Pi_h \neq I$  on the continuum space. This is ordinary representation loss; it is not by itself a probability law.

The label exchanges  $G \leftrightarrow E$  and  $D \leftrightarrow C$  form a formal  $\mathbb{Z}_2 \times \mathbb{Z}_2$  action on labels. A physical symmetry requires value-space identifications and invariance of both action and residual. TQC III is correct to withhold that claim.

### 3 Bridge-controlled closure theorem

For  $u, v \in \mathbb{R}^6$  and a  $6 \times 6$  permutation matrix  $P$ , define

$$F_P(u, v) = \begin{pmatrix} I & I \\ P & I \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} u + v \\ v + Pu \end{pmatrix}.$$

The first block gives  $v = -u$ ; the second gives  $(P - I)u = 0$ . Therefore

$$\ker F_P = \{(u, -u) : Pu = u\}, \quad \dim \ker F_P = \dim \text{Fix}(P).$$

For a permutation, the fixed-vector dimension equals its number of disjoint cycles. Identity, one transposition, and one six-cycle therefore have nullities 6, 5, and 1, respectively. Figure 1 shows the corresponding rank/nullity tradeoff.

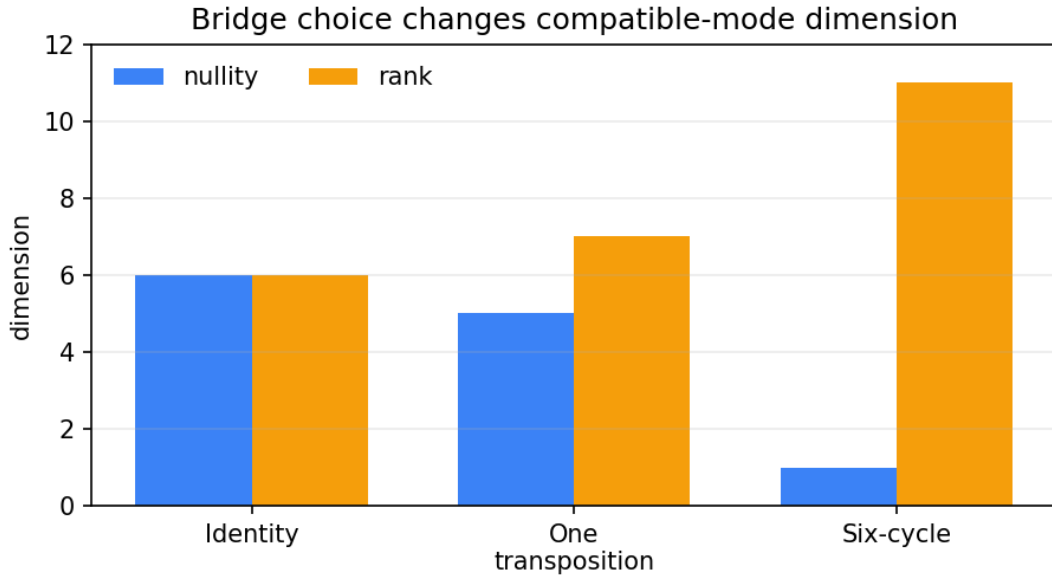


Figure 1: The same twelve descriptive slots support different compatible-mode dimensions when the bridge changes. This is a classical linear constraint result.

This theorem establishes bridge-controlled mode selection. It does not identify the modes with particles, fields, or quantum states. The relaxation iteration used in the note is an optimizer: its iteration index is not physical time, and its dissipative convergence is not reversible quantum evolution.

### 4 Phase and representation consistency

If a continuous map  $A : \mathbb{R} \rightarrow U(1)$  obeys

$$A(S_1 + S_2) = A(S_1)A(S_2),$$

then  $A(S) = e^{i\alpha S}$ , with  $\alpha = 1/\kappa$  a convention-dependent action scale. This is a correct conditional theorem. It assumes the phase group, continuity, and action additivity.

For a finite history kernel

$$K_{ba} = \sum_{h:a \rightarrow b} w_h e^{iS[h]/\kappa},$$

unit-modulus factors alone do not imply  $K^\dagger K = I$ . A readout of the form

$$M_b(\psi) = \frac{|\psi_b|^2}{\sum_c |\psi_c|^2}$$

is a Born-form assumption, not a consequence of closure.

The paired-real representation makes the control explicit. If  $K = A + iB$ , then

$$\mathcal{R}(K) = \begin{pmatrix} A & -B \\ B & A \end{pmatrix}, \quad \mathcal{R}(\psi) = \begin{pmatrix} \Re\psi \\ \Im\psi \end{pmatrix},$$

and  $\mathcal{R}(K\psi) = \mathcal{R}(K)\mathcal{R}(\psi)$ . A complex calculation and a structured paired-real calculation can therefore be exactly equivalent.

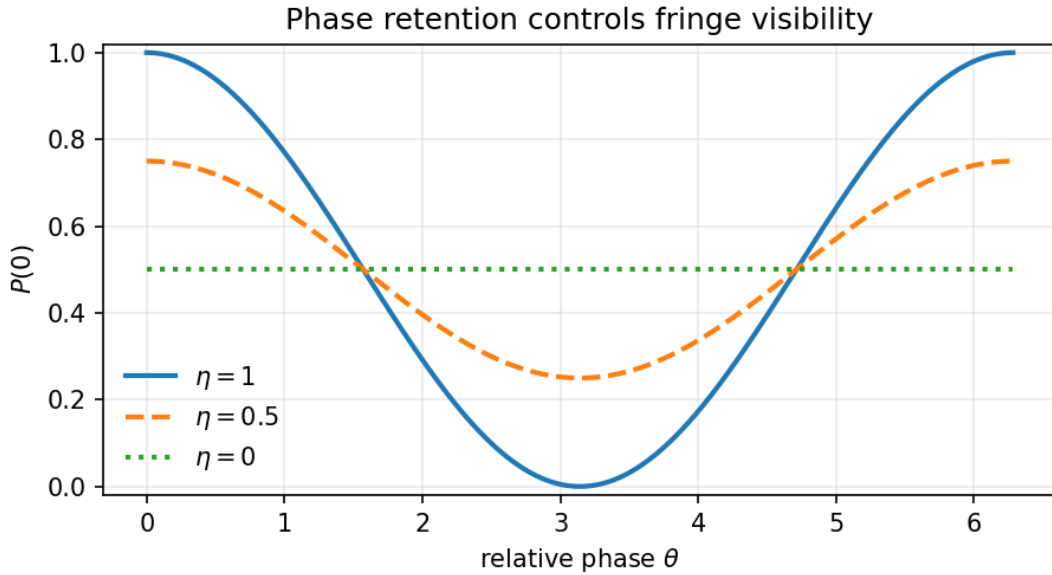


Figure 2: Two-channel interference under an explicitly chosen unitary mixer.  $\eta$  is imposed coherence; the plot verifies the model and does not derive the model from gravity or electromagnetism.

## 5 Triangulated gauge and Regge control

We implemented a square domain split into counter-clockwise right triangles. Flat Euclidean edge lengths give zero interior two-dimensional Regge deficits. On each oriented edge we integrate the smooth connection

$$A = (-By/2, Bx/2),$$

and sum edge links around each triangle. A vertex gauge transformation adds the coboundary  $d_0\chi$ ; triangle flux is unchanged because  $d_1d_0 = 0$ .

For mesh spacing  $h = 1/n$ , the compact Wilson action is scaled by  $n^2$  and approaches the constant-field continuum target  $B^2/4 = 0.25$ . The benchmark results are:

| $n$ | Wilson action | absolute error        | max. deficit |
|-----|---------------|-----------------------|--------------|
| 2   | 0.24967465    | $3.25 \times 10^{-4}$ | 0            |
| 4   | 0.24997966    | $2.03 \times 10^{-5}$ | 0            |
| 8   | 0.24999873    | $1.27 \times 10^{-6}$ | 0            |
| 16  | 0.24999992    | $7.95 \times 10^{-8}$ | 0            |
| 32  | 0.25000000    | $4.89 \times 10^{-9}$ | 0            |
| 64  | 0.25000000    | $< 10^{-15}$          | 0            |

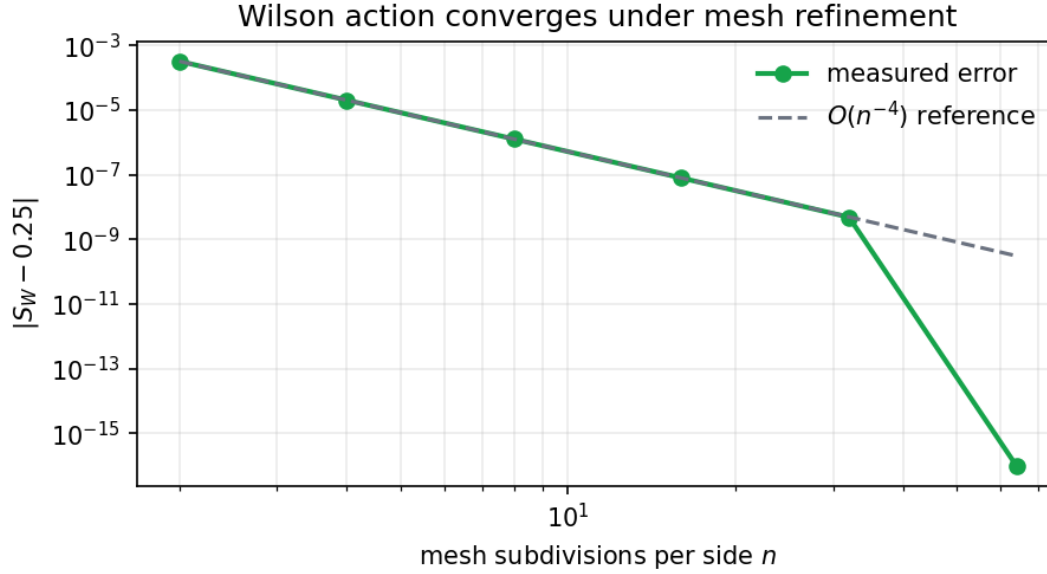


Figure 3: Wilson-action refinement. The observed fourth-order error is consistent with the exact line integration and cosine expansion used in this smooth constant-field test.

This is a genuine discrete gauge and refinement verification. It is not yet a physical Einstein–Maxwell simulation: the geometry is flat, there is no matter stress tensor, and no coupled dynamical evolution is solved.

## 6 Assessment

| Claim or mechanism                                                            | Status                                                |
|-------------------------------------------------------------------------------|-------------------------------------------------------|
| Twelve descriptive roles need not be twelve physical dimensions               | Correctly formalized by typed spaces and bridge maps. |
| Bridge choice can change compatible-mode dimension                            | Proven in the finite permutation model.               |
| Continuous additive action plus $U(1)$ multiplication gives exponential phase | Correct conditional theorem.                          |
| Phase factors imply reversible dynamics                                       | False without a unitary kernel condition.             |

|                                                                          |                                                      |
|--------------------------------------------------------------------------|------------------------------------------------------|
| Coarse-graining creates quantum probabilities                            | False; readout and measure must be specified.        |
| The triangle EM identity demonstrates a unified theory                   | False; it verifies a discrete cochain identity only. |
| The present work derives quantum mechanics from Einstein–Maxwell closure | Not established.                                     |

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The strongest defensible conclusion is:

A typed discrete–continuous bridge can constrain compatible modes and can preserve or erase relative-phase information. Under explicitly adopted phase and readout rules, the resulting finite model supports interference, with an equivalent paired-real control.

## 7 Decisive next test

Use one non-flat triangulated mesh with Regge edge lengths, compact gauge links, fixed boundary data, a specified matter action, and a refinement sequence. Compare positive-real, signed-real, complex-phase, and paired-real models at matched parameter counts. Record typed residuals, gauge covariance, round-trip bridge error, evolution-commutation defect, unitarity error, held-out observables, runtime, and memory. The complex model earns explanatory value only if the bridge yields a held-out restriction or convergence advantage that was not inserted into its definition.

## 8 Reproducibility

The source review is in `TQC_III_EVALUATION.md`. The triangulated implementation is in `src/triangulated_closure.py` run:

```
cd /home/rmichaelglover/relaxation-publication-study
python3 paper/make_tqc3_figures.py
PYTHONPATH=. python3 benchmark_triangulated.py
PYTHONPATH=. python3 -m unittest discover -s tests -v
```

The full project suite passes 17 tests. Synthetic numerical agreement measures agreement with the stated mathematics, not agreement with nature.

## References

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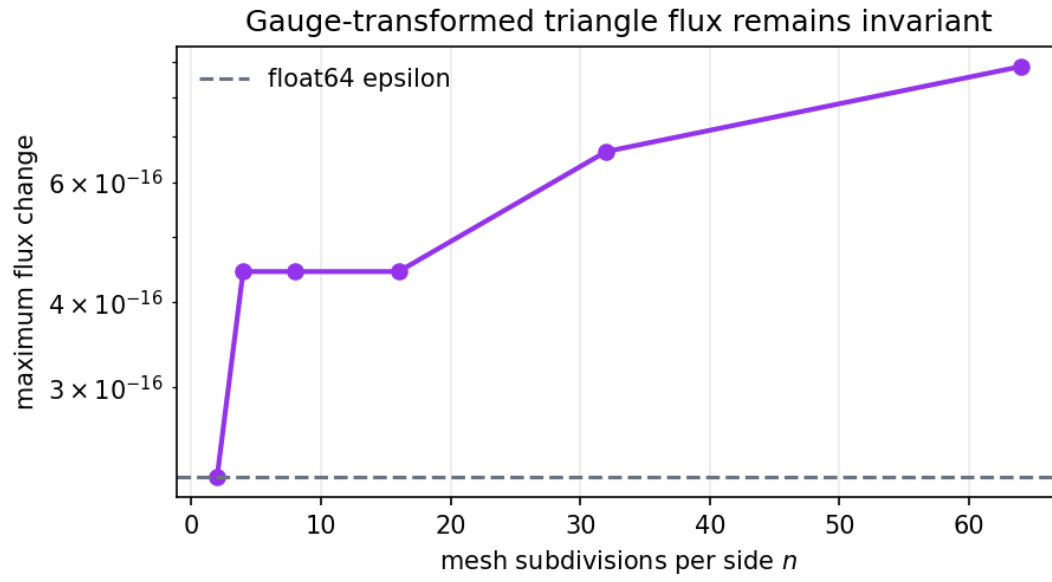


Figure 4: Maximum change in triangle flux after random vertex gauge transformations. Errors remain at floating-point roundoff.